HYDROCHLOROTHIAZIDE

Morning dose

Hyponatraemia, hypokalaemia

CHEMICAL NAME 6-chloro-1,1-dioxo-3,4-dihydro-2H-1λ⁶,2,4-benzothiadiazine-7-sulfonamide

CLASS Thiazide diuretic

TRADE NAMES Esidrex · HydroDiuril · Microzide · Combinations: Co-Diovan, Hyzaar

THERAPEUTIC USES / INDICATIONS

- Hypertension (alone or in combination).
- Mild oedema (heart failure, cirrhosis).
- Nephrogenic diabetes insipidus.
- Prevention of calcium kidney stones (hypercalciuria).

ADMINISTRATION

- **PO** once daily in the morning.
- Often in fixed-dose combination with ACEi/ARB.

DOSAGES

- **Hypertension:** 12.5–25 mg once daily (max 50 mg).
- **Oedema:** 25–100 mg daily.
- **Children:** 1–2 mg/kg/day.
- Less effective if CrCl <30 mL/min.

PHARMACOKINETICS

- **Onset:** 2 h. **Peak:** 4 h. **Duration:** 6–12 h.
- Not metabolised.
- Excreted unchanged in urine. **t½:** 6–15 h.

PHARMACODYNAMICS

- Inhibits **Na^+/Cl^- co-transporter** in distal convoluted tubule → moderate natriuresis.
- ↓ Ca^{2+} excretion (hypercalcaemia); ↑ K^+ and Mg^{2+} loss.
- Long-term BP lowering via ↓ peripheral resistance.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Increased urination, dizziness.
- Postural hypotension.
- Photosensitivity.

ADVERSE EFFECTS (SERIOUS)

- Hyponatraemia (elderly women), hypokalaemia, hypomagnesaemia, hypercalcaemia.
- Hyperuricaemia → gout.
- Hyperglycaemia, dyslipidaemia.
- Non-melanoma skin cancer (long-term), acute myopia/angle-closure glaucoma, pancreatitis.

CONTRAINDICATIONS & CAUTIONS

- Anuria; severe renal impairment (ineffective).
- Refractory hypokalaemia, hyponatraemia, hypercalcaemia.
- Symptomatic hyperuricaemia/gout.
- Sulfonamide hypersensitivity (caution); Addison's disease.

KEY INTERACTIONS

- **Lithium:** ↑ lithium levels.
- **Digoxin:** hypokalaemia → toxicity.
- **NSAIDs:** ↓ effect.
- **Antidiabetics:** ↓ glucose control.

NURSING MANAGEMENT

- Monitor BP, U&E (Na^+, K^+), glucose, uric acid.
- Watch elderly for confusion (hyponatraemia).
- Assess for gout attacks.
- Daily weight if used for oedema.

PATIENT EDUCATION

- Take in the morning.
- Use sunscreen; check skin regularly.
- Report muscle cramps, weakness, confusion, joint pain.
- Rise slowly.

KEY POINTS

- Thiazide — blocks NaCl co-transport in DCT.
- Raises uric acid and glucose.
- Hyponatraemia, hypokalaemia, hypercalcaemia.
- Ineffective in severe renal failure.